



UNIT-III

SQL Commands

Structured Query Language commands are commonly grouped into categories:

- 1.) DDL — It stands for Data Definition Language.
 - Used to define & modify database structures.
- a.) Create — Create a new table or database.
- b.) Alter — Modifies the structure of an existing table.
- c.) Drop — Permanently deletes a table.
- d.) Truncate — Removes all rows from a table but keeps the table structure.
- e.) Rename — Changes the name of a table.
- 2.) DML — It stands for Data Manipulation Language
 - Used to manage data in tables.

Summary :



- a) Insert — Adds new records.
- b) Update — Modifies existing records.
- c) Delete — Delete specific records
- 3.) DQL — • It stands for Data Query Language.
 - Use to retrieve data.
- a) Select — Display data from a table.
- 4.) DCL — • It stands for Data Control Language.
 - Used to control access.
- a) Grant — Gives permissions to a user.
- b) Revoke — Remove permissions.
- 5.) TCL — • It stands for Transaction Control Language.
 - Used to manage transactions.
- a) Commit — Saves all changes permanently
- b) Rollback — Cancels changes made after the last COMMIT.

Summary :



⇒ Savepoint — Creates a checkpoint within a transaction.

Mnemonic: “Don't Make Queries During Transactions”

D → Define ⇒ DDL

M → Modify Data ⇒ DML

Q → Query Data ⇒ DQL

D → Data Control ⇒ DCL

T → Transaction ⇒ TCL

Views in SQL

A View^o is a virtual table created from one or more existing tables using a SELECT statement. It stores the query, not the actual data.

⇒ A view^o is a virtual table based on the result of a SQL query.

```
CREATE VIEW view_name AS
SELECT column1, column2
FROM table_name
WHERE condition;
```

Summary :



Ex:-

Student Table:

ID	Name	Age
1	Rahul	21
2	Priya	19

Create a view:

```
CREATE VIEW AdultStudents AS  
SELECT ID, Name  
FROM Student  
WHERE Age >= 20;
```

Display the view:

```
SELECT * FROM AdultStudents;
```

Output:

ID	Name
1	Rahul

* is a wildcard symbol used to select all columns from a table.

In SQL, the * (asterisk) means "all columns".

Summary :



DDL Commands

CREATE Syntax

```
CREATE TABLE student (  
    ID INT,  
    Name VARCHAR(50)  
);
```

ALTER Syntax

```
ALTER TABLE student  
ADD Age INT;
```

DROP Syntax

```
DROP TABLE student;
```

TRUNCATE Syntax

```
TRUNCATE TABLE student;
```

RENAME Syntax

```
ALTER TABLE student  
RENAME TO students;
```

DML Commands

INSERT Syntax

```
INSERT INTO student  
VALUES (1, 'Rahul', 20);
```

Summary :



UPDATE Syntax

```
UPDATE student  
SET Age = 21  
WHERE ID = 1;
```

DELETE Syntax

```
DELETE FROM student  
WHERE ID = 1;
```

DQL Commands

SELECT Syntax

```
SELECT * FROM student;
```

DCL Commands

GRANT Syntax

```
GRANT SELECT  
ON student  
TO User1;
```

REVOKE Syntax

```
REVOKE SELECT  
ON student  
FROM User1;
```

Summary :



TCL Commands

COMMIT Syntax

COMMIT;

ROLLBACK Syntax

ROLLBACK;

SAVEPOINT Syntax

SAVEPOINT;

Query Processing & Query Optimization

Query Processing is the process by which a DBMS receives a SQL query, understands it, executes it, & returns the result.

Query Processing is a procedure of transforming a high-level query (such as SQL) into a correct & efficient execution plan expressed in low-level language. A query processing selects a most appropriate plan that is used in responding to a database request. When a database system receives a query for update or retrieval of information, it goes through a series of compilation steps, called execution plan.

Summary :

Query Optimization is the process of finding the fastest & most efficient way to execute a query.

Query Processing Steps :-

1.) Receive the Query
The DBMS receives the SQL query from the user.

2.) Check the Query

The DBMS checks the query for syntax errors, table names, column names & verifies whether the query is valid.

3.) Process & Execute the Query

If the query is correct, the DBMS processes it & accesses the required data from the database.

4.) Retrieve the Data

The DBMS fetches the required records from the database.

5.) Return the Result

Finally, the DBMS displays the result to the user.

Query Optimization Steps :-

1.) Receive the Query

The DBMS receives the SQL query from the user.

2.) Analyze the Query

The query optimizer examines the query & identifies different possible execution plans.

Summary :



3.) Choose the Best Execution Plan

The optimizer compares all possible plan & selects the one with the lowest cost, least resource usage & fastest execution time.

4.) Execute the Query

The DBMS executes the selected execution plan to retrieve the required data.

5.) Return the Result

Finally, the DBMS displays the retrieved data to the user.

Query Cost-

Query cost in a Database Management System (DBMS) is an internal estimate of the total resource usage required to execute a specific database query plan. Before executing a query, the system's query optimizer evaluates multiple possible execution plans & calculates their respective costs to select the most efficient path.

Key Components of Query Cost — The overall cost of a query is an accumulation of several physical resource metrics:

Summary :



Measures of Query Cost -

1.) Disk Access Cost - Disk access is the most expensive operation in DBMS.

- Cost of reading / writing data from disk.

It includes :

- Reading data from disk
- Writing data to disk

Ex:- If a query reads 1000 blocks from disk, the cost is higher ~~that~~ than reading 10 blocks.

2.) CPU Cost - CPU cost is the processing time required to execute a query.

- Processing time required for computations.

CPU performs calculations & comparisons.

More comparisons = Higher CPU Cost.

Summary :



3.) Memory Cost — Memory required during query execution.

- Amount of memory used during execution.

More memory usage = Higher Cost.

4.) Communication Cost — Important in distributed databases.

- Cost of transferring data across a network.
- It is the cost of transferring data over a network.

5.) Response Time — Time taken to produce the result.

- Total time taken to ~~send~~ return the result.

Ex:- Query A = 2 secs

Query B = 10 secs.

Query A has lower response time & lower cost.

Summary :

Selection Operation (σ)

Selection operation is used to select rows (tuples) from a relation (table) that satisfy a given condition.

Syntax:

$\sigma_{\text{condition}}(\text{Relation})$

Ex:

Student Table

Roll No	Name	Dept
101	Amit	IT
102	Neha	CSE
103	Rahul	IT

Query:

$\sigma_{\text{Dept} = 'IT'}(\text{Student})$

Result:

Roll no	Name	Dept
101	Amit	IT
103	Rahul	IT

Types of Selection Operation

- 1.) Simple Selection — Uses a single condition to select tuples.
- 2.) Conjunctive Selection (AND) — Uses two or more conditions connected by ~~AND~~ AND ($\wedge$).
- 3.) Disjunctive Selection (OR) — Uses conditions connected by OR ($\vee$).
- 4.) Negation Selection (NOT) — Uses the NOT ($\neg$) operator to exclude tuples satisfying a condition.
- 5.) Compound Selection — Uses a combination of AND, OR & NOT operators.

Sorting Operations

Sorting is the process of arranging records (rows) of a table in a specific order based on one or more attributes (columns). The order can be ascending (ASC) or descending (DESC).

Summary :



Syntax

SELECT * FROM Student
ORDER BY Name ASC;

OR

SELECT * FROM Student
ORDER BY Marks DESC;

Types of Sorting

- 1.) Ascending Order (ASC) - Smallest to largest, A to Z.
- 2.) Descending Order (DESC) - Largest to smallest, Z to A.

Join Operation

A Join operation combines rows from two or more tables based on a related attribute (common column).

Types of Join Operation

- 1.) Inner Join - Returns only matching rows from both tables.

Summary :

Ex:- Student-Table

RollNo	Name
101	Amit
102	Neha

Course-Table

RollNo	Course
101	DBMS
103	OS

Query:-

SELECT * FROM student

INNER JOIN Result

ON Student. RollNo. = Result. RollNo;

Output:-

RollNo.	Name	Course
101	Amit	DBMS

- 2.) Left Outer Join — Returns all rows from the left table & matching rows from the right table.
- 3.) Right Outer Join — Return all rows from the right table & matching rows from the left table.
- 4.) Full Outer Join — Return all matching & non-matching rows from both tables.

Summary :



5.) Natural Join — Automatically joins tables based on columns having the same name & data types.

Ex:-

```
SELECT * FROM Student  
NATURAL JOIN Result;
```

6.) Self Join — A table is joined with itself.

Ex:-

```
SELECT E1.Name, E2.Name  
FROM Employee E1, Employee E2  
WHERE E1.ManagerID = E2.EmpID;
```

7.) Cross Join — Return all possible combinations of rows from both tables.

Ex:-

```
SELECT * FROM Student  
CROSS JOIN Course;
```

If student has 3 rows & Course has 4 rows, the result will contain $3 \times 4 = 12$ rows.

Summary :



Evaluation of Expressions

Evaluation of an expression is the process of executing a relational algebra query & producing the required result.

The DBMS converts a query into an execution plan & performs operations such as selection, projection, join & sorting to obtain the final output.

Ex:- Query:

```
SELECT Name  
FROM student  
WHERE Dept = 'IT';
```

Relational Algebra:

$\pi_{Name} (\sigma_{Dept = 'IT'} (Student))$

Steps in Expression Evaluation

- 1.) Read the query.
- 2.) Convert SQL into relational algebra.
- 3.) Choose an efficient execution strategy.

Summary :

- 4.) Execute relational operations.
- 5.) Return the final result.

Importance :-

- Reduces execution time.

- Improves query performance.

- Minimizes resource usage.

Transformation of Relational Expressions

Transformation of relational expressions is the process of converting a relational algebra expressions into an equivalent but more efficient expression.

The result remains the same, but execution becomes faster.

Common Transformation Rules

1.) Selection Cascade

$$\sigma_{C_1 \wedge C_2}(R) = \sigma_{C_1}(\sigma_{C_2}(R))$$

2.) Commutative Property of Selection

$$\sigma_{C_1}(\sigma_{C_2}(R)) = \sigma_{C_2}(\sigma_{C_1}(R))$$

Summary :

3.) Projection Cascade

$$\pi_{L_1}(\pi_{L_2}(R)) = \pi_{L_1}(R)$$

4.) Selection Push Down

Move selections as close as possible to base relations.

Advantages • faster execution

- Less disk I/O.
- Better query optimization.

Estimating Statistics of Expression Results

Estimating statistics means predicting the size & characteristics of intermediate query results before executing the query.

The optimizer uses these estimates to choose the best execution plan.

Statistics Maintained by DBMS

- 1.) Number of tuples
- 2.) Number of blocks
- 3.) Distinct values

Summary :



4.) Attribute distribution

5.) Selectivity factor

Evaluation Plan

An evaluation plan is a sequence of operations used by the DBMS to execute a query.

It specifies:

- Which operation will be executed first.
- Which access method will be used.
- Which join algorithm will be chosen.

Ex^o Query:

```
SELECT * FROM Student S, Result R  
WHERE S.RollNo = R.RollNo;
```

Possible Plans:

- Plan 1.
- a.) Read Student
 - b.) Read Result
 - c.) Perform Join
 - d.) Display Result

Summary :

- Plan 2
- Use index on Roll No.
 - Retrieve matching tuples
 - Perform indexed join
 - Display Result.

Plan 2 is usually faster.

Types of Evaluation Plan

- 1.) Naive Evaluation Plan
Direct execution without optimization.
- 2.) Optimized Evaluation Plan
Uses transformation rules & statistics to reduce cost.
- 3.) Indexed Evaluation Plan
Uses index for faster retrieval.
- 4.) Join-Based Evaluation Plan
Chooses the best join algorithm such as:
 - Nested Loop Join
 - Merge Join
 - Hash Join

Summary :



Case study of Oracle & db2

Oracle Database — Oracle Corporation developed Oracle

Database, one of the most widely used Relational Database Management Systems (RDBMS). It is used by large organizations for storing & managing huge amounts of data.

Features of Oracle Database

- Supports SQL & PL/SQL.
- High security & data protection.
- Backup & recovery facilities.
- Concurrency control & transaction management.
- Scalability for large databases.
- Distributed database support.

Applications

- Banking systems
- E-commerce websites
- Government organizations
- Healthcare systems

Summary :



- Excellent backup & recovery.

Limitations

- Expensive licensing cost
- Requires skilled administrators
- Complex configuration

IBM DB2 — IBM developed DB2, a powerful RDBMS designed for managing structured data efficiently. It is commonly used in enterprise-level applications.

Features of DB2

- Supports SQL standards.
- Advanced query optimization.
- High availability & reliability.
- Data compression techniques.
- Strong security features.
- Supports distributed databases.

Applications

- Banking & finance

Summary :



- Insurance companies
- Retail businesses
- Telecommunications
- Data warehousing

Advantages

- ~~Exe~~ Excellent performance
- Efficient storage management
- Strong security
- Good scalability

Limitations

- Licensing cost is high
- Learning curve for beginners
- Less popular than Oracle in some sectors.

Summary :